EDA for the road features extracted from map image

Makoto Hanyuda (24773603)

2026-07-27

Preparing the Analysis Dataset

This section merges the pedestrian survey data with the road feature data collected from images. The response variables are `obs_peak_along` and `obs_peak_cross`, representing observed pedestrian counts along and across the road. The explanatory variables include road and land-use features such as footpath type, building counts, and pedestrian infrastructure.

```
# Load data
library(fst)
df <- read_fst("data/Ped_iRAP_merged.fst")
road_features <- read.csv("data/ausrap_with_gemini_features.csv", stringsAsFactors = FALSE)

# Convert True/False character columns to logical
bool_cols <- c(
  "footpath_both_sides", "footpath_one_side", "fencing_barriers",
  "railway_station", "bus_stops", "overpass_bridge",
  "park_bushland_reserve", "shared_pedestrian_and_cycle_path"
)
for (col in bool_cols) {
  road_features[[col]] <- road_features[[col]] == "True"
}

# Rename keys and merge
names(road_features)[names(road_features) == "road_no"] <- "ROAD_NO"
names(road_features)[names(road_features) == "slk_km"] <- "SLK"

merged <- merge(df, road_features, by = c("ROAD_NO", "SLK"))

# Create footpath category
merged$footpath_cat <- factor(
  ifelse(merged$footpath_both_sides & merged$footpath_one_side, "footpath_mixed",
  ifelse(merged$footpath_both_sides, "footpath_both_sides",
  ifelse(merged$footpath_one_side, "footpath_one_side", "footpath_none"))),
  levels = c("footpath_none", "footpath_one_side", "footpath_mixed", "footpath_both_sides")
)

response_vars <- c("obs_peak_along", "obs_peak_cross")

predictor_vars <- c(
  "footpath_cat", "pedestrian_crossings", "commercial_buildings",
  "residential_buildings", "industrial_buildings", "parking_areas",
  "fencing_barriers", "railway_station", "bus_stops", "overpass_bridge",
  "park_bushland_reserve", "shared_pedestrian_and_cycle_path"
```

```
)

analysis_df <- merged[, c(response_vars, predictor_vars)]
str(analysis_df)
```

```
## 'data.frame':    46 obs. of  14 variables:
##  $ obs_peak_along                 : num  37.33 137.33 14.33 4.67 11.67 ...
##  $ obs_peak_cross                 : num  18.67 107.33 3 11.33 3.33 ...
##  $ footpath_cat                   : Factor w/ 4 levels "footpath_none",..: 2 2 2 2 2 3 2 1 2 2 ...
##  $ pedestrian_crossings           : int  1 1 0 0 0 0 0 0 0 1 ...
##  $ commercial_buildings           : int  3 2 2 1 2 0 0 0 2 1 ...
##  $ residential_buildings          : int  2 0 6 2 5 2 0 0 10 5 ...
##  $ industrial_buildings           : int  0 0 0 0 0 0 1 0 1 0 ...
##  $ parking_areas                  : int  1 2 2 1 0 1 1 0 2 2 ...
##  $ fencing_barriers               : logi  TRUE TRUE TRUE TRUE TRUE TRUE ...
##  $ railway_station                : logi  FALSE FALSE FALSE FALSE FALSE FALSE ...
##  $ bus_stops                      : logi  FALSE FALSE FALSE FALSE FALSE FALSE ...
##  $ overpass_bridge                : logi  FALSE FALSE FALSE FALSE FALSE FALSE ...
##  $ park_bushland_reserve          : logi  TRUE TRUE FALSE TRUE TRUE TRUE ...
##  $ shared_pedestrian_and_cycle_path: logi  TRUE TRUE FALSE TRUE FALSE FALSE ...
```

```
table(analysis_df$footpath_cat)
```

```
##
##       footpath_none   footpath_one_side      footpath_mixed footpath_both_sides
##                   6                  23                   8                   9
```

Correlation Check: Numeric Variables

Before fitting boxplots for the categorical variables, this section checks the numeric explanatory variables against both response variables. Rows with missing values are removed to keep the correlation calculation valid.

```
numeric_vars <- c(
  "pedestrian_crossings", "commercial_buildings",
  "residential_buildings", "industrial_buildings", "parking_areas"
)

num_data <- analysis_df[, c("obs_peak_along", "obs_peak_cross", numeric_vars)]
num_data <- num_data[complete.cases(num_data), ]
summary(num_data)
```

```
##  obs_peak_along     obs_peak_cross    pedestrian_crossings commercial_buildings
##  Min.   :  0.6667   Min.   :  0.000   Min.   :0.0000       Min.   : 0.000
##  1st Qu.:  8.1667   1st Qu.:  3.333   1st Qu.:0.0000       1st Qu.: 0.000
##  Median : 13.6667   Median :  7.833   Median :0.0000       Median : 2.000
##  Mean   : 41.0036   Mean   : 26.594   Mean   :0.3043       Mean   : 2.804
##  3rd Qu.: 34.1667   3rd Qu.: 32.250   3rd Qu.:0.0000       3rd Qu.: 4.000
##  Max.   :349.6667   Max.   :192.333   Max.   :4.0000       Max.   :12.000
##  residential_buildings industrial_buildings parking_areas
##  Min.   : 0.00         Min.   :0.0000       Min.   :0.00
##  1st Qu.: 0.00         1st Qu.:0.0000       1st Qu.:1.00
##  Median : 2.50         Median :0.0000       Median :2.00
##  Mean   : 3.63         Mean   :0.1739       Mean   :1.63
##  3rd Qu.: 6.00         3rd Qu.:0.0000       3rd Qu.:2.00
```

```
##  Max.   :16.00          Max.   :3.0000        Max.   :4.00
```

```
library(PerformanceAnalytics)
```

```
## Loading required package: xts
```

```
## Loading required package: zoo
```

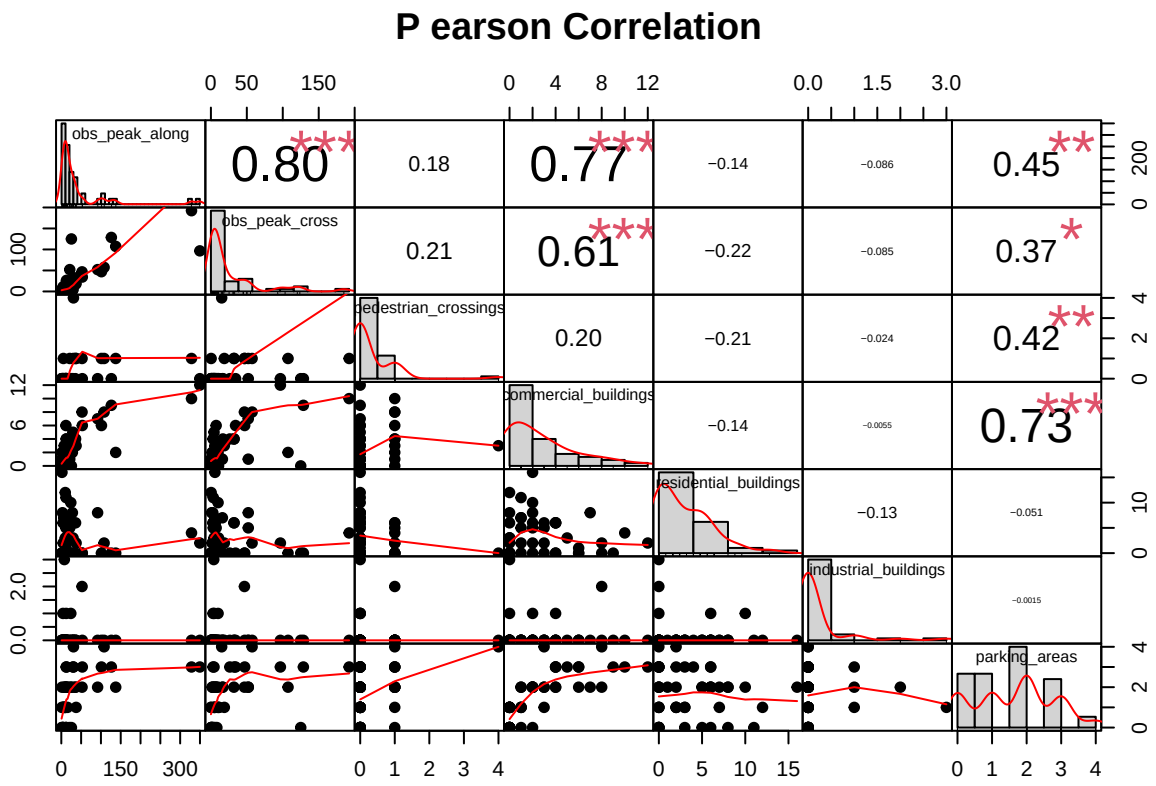
```
##
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```
##
## Attaching package: 'PerformanceAnalytics'
```

```
## The following object is masked from 'package:graphics':
##
##     legend
```

```
chart.Correlation(num_data, histogram = TRUE, pch = 19)
```



Variables significant for obs_peak_along

- commercial_buildings (r = 0.77, ***)
- parking_areas (r = 0.45, **)

Variables significant for obs_peak_cross

- commercial_buildings (r = 0.61, ***)
- parking_areas (r = 0.37, *)

Both target variables show significant correlation with the same two predictors: `commercial_buildings` and `parking_areas`. The correlation is stronger for `obs_peak_along` than for `obs_peak_cross`.

Count of TRUE/FALSE for Categorical Variables

This section checks the number of TRUE and FALSE observations for each binary explanatory variable (excluding `footpath_cat`), to identify variables with insufficient sample size in either group.

```
cat_vars_check <- c(
  "fencing_barriers", "railway_station", "bus_stops",
  "overpass_bridge", "park_bushland_reserve", "shared_pedestrian_and_cycle_path"
)

count_table <- data.frame(
  variable = cat_vars_check,
  FALSE_n  = sapply(cat_vars_check, function(v) sum(analysis_df[[v]] == FALSE, na.rm = TRUE)),
  TRUE_n   = sapply(cat_vars_check, function(v) sum(analysis_df[[v]] == TRUE, na.rm = TRUE))
)

knitr::kable(count_table, caption = "TRUE/FALSE counts by categorical variable")
```

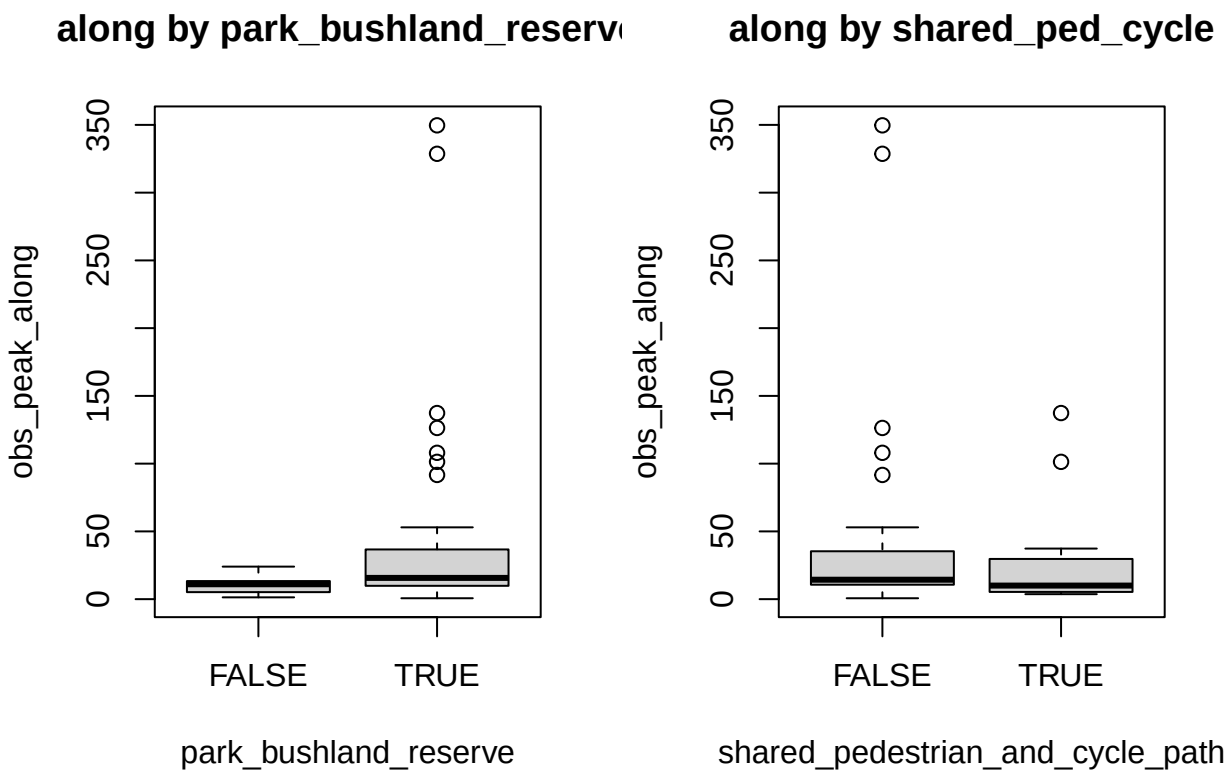
Table 1: TRUE/FALSE counts by categorical variable

	variable	FALSE_n	TRUE_n
fencing_barriers	fencing_barriers	3	43
railway_station	railway_station	44	2
bus_stops	bus_stops	44	2
overpass_bridge	overpass_bridge	45	1
park_bushland_reserve	park_bushland_reserve	7	39
shared_pedestrian_and_cycle_path	shared_pedestrian_and_cycle_path	33	13

Note on Variable Selection

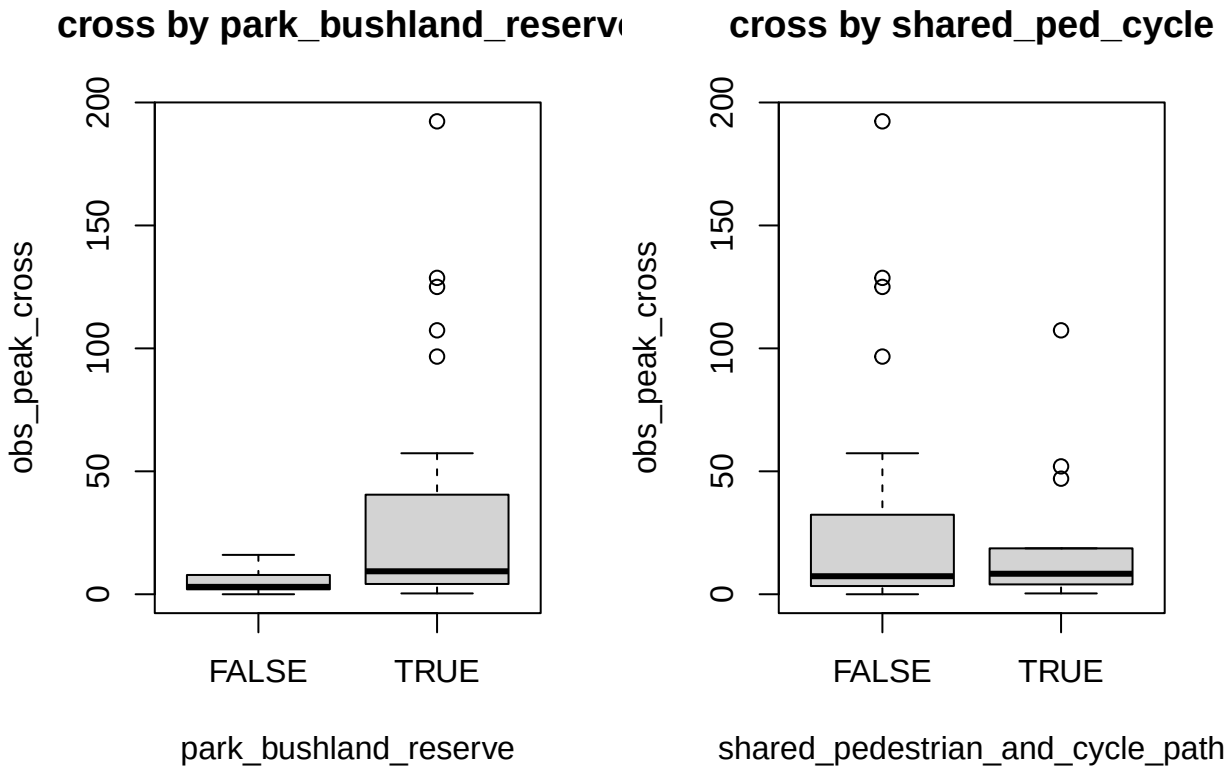
Based on the TRUE/FALSE counts above, `bus_stops` and `overpass_bridge` have 2 or fewer TRUE cases, and `fencing_barriers` has only 3 FALSE cases. These imbalances make the comparisons unreliable, so these three variables are excluded from further analysis. The following boxplots therefore focus on `park_bushland_reserve` and `shared_pedestrian_and_cycle_path`, which have a more balanced group size.

```
par(mfrow = c(1, 2))
boxplot(merged$obs_peak_along ~ merged$park_bushland_reserve,
        xlab = "park_bushland_reserve", ylab = "obs_peak_along",
        main = "along by park_bushland_reserve")
boxplot(merged$obs_peak_along ~ merged$shared_pedestrian_and_cycle_path,
        xlab = "shared_pedestrian_and_cycle_path", ylab = "obs_peak_along",
        main = "along by shared_ped_cycle")
```




```
par(mfrow = c(1, 1))
```

```
par(mfrow = c(1, 2))
boxplot(merged$obs_peak_cross ~ merged$park_bushland_reserve,
        xlab = "park_bushland_reserve", ylab = "obs_peak_cross",
        main = "cross by park_bushland_reserve")
boxplot(merged$obs_peak_cross ~ merged$shared_pedestrian_and_cycle_path,
        xlab = "shared_pedestrian_and_cycle_path", ylab = "obs_peak_cross",
        main = "cross by shared_ped_cycle")
```




```
par(mfrow = c(1, 1))
```

Summary of Boxplot Findings

park_bushland_reserve - For both `obs_peak_along` and `obs_peak_cross`, the TRUE group shows a higher median and wider spread than the FALSE group, including several high outliers. - The FALSE group has only 7 observations, so this comparison should be interpreted with caution.

shared_pedestrian_and_cycle_path - For both targets, the FALSE group shows a slightly higher median than the TRUE group, but the boxes overlap considerably. - The difference between the two groups is not clearly distinguishable from this plot alone.

Overall - `park_bushland_reserve` shows a more visible difference between groups than `shared_pedestrian_and_cycle_path` for both target variables.

Conclusion

`commercial_buildings` and `parking_areas` had the strongest correlation with both target variables. `footpath_cat` was also significant, but it may just reflect the effect of `commercial_buildings`. Four binary variables had too few cases in one group to be reliable, so they were removed. Of the remaining two, `park_bushland_reserve` showed a clear difference between groups, but `shared_pedestrian_and_cycle_path` did not.

For the next stage, `commercial_buildings` and `parking_areas` are the strongest candidates. `footpath_cat` and `park_bushland_reserve` need a multicollinearity check before they can be used.